

chatApp

A small overlay chat system features:

Online directory (user list)

Private messages (point-to-point)

Group messages (public channel / broadcast)

Point-to-point file transfer (chunked)

Secure coding: timestamp window, replay protection, content signatures

Ethical backdoors (documented at the end) to support peer review & hackathon

Server = Node.js (Express + ws), Client = pure web (WebCrypto).

Default port: 8080, static client is served from /client.

1) Quick start

Requirements:

Node.js  $\geq$  18 (tested with v18 & v20)

Browsers with WebCrypto (Chrome/Edge/Safari recent)

Run:(cd chatApp)

npm start

2) Project structure

chatApp/

├─ server.js

├─ package.json

├─ client/

| └─ index.html

| └─ styles.css

| └─ app.js

| └─ b64url.js

| └─ crypto.js

└─ README.md

### 3) How to use

Open <http://localhost:8080> in two browser windows or two machines.

Enter a nickname and click Join.

The Online list panel shows current peers (id + stable id).

Pick a recipient and type a message → Send.

Send to all posts to the public channel.

Send file → choose a file to send point-to-point (chunked).

### 4) Contact:

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